

SURFACE AREA

any part of the outside of a shape that you can touch.

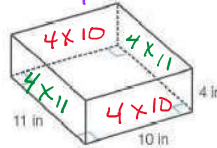
Surface area is the sum of the areas of the faces of a three-dimensional figure. You already know how to find the area of a complex figure. You can use this knowledge to find surface area.

FINDING SURFACE AREA BY ADDING AREAS

Example Find the surface area of the box.

The box, a rectangular prism, has 6 faces. Each face is a rectangle that has its own length and width.

Imagine unfolding the box so that the cardboard lies flat. Make sure you understand how the second drawing can be folded to make the box in the first drawing.



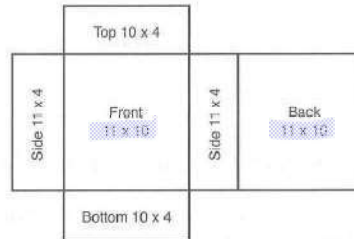
Step 1

Find the area of each rectangle.

Top:	$10 \times 4 = 40$	40
Bottom:	$10 \times 4 = 40$	40
Left side:	$11 \times 4 = 44$	44
Right side:	$11 \times 4 = 44$	44
Front:	$11 \times 10 = 110$	110
Back:	$11 \times 10 = 110$	110
		<u>+ 110</u>
		388

Step 2

Add the areas of the faces.



The surface area of the box is **388 square inches**.

a Closed Box

The Surface Area SA is the sum of the areas of the top, bottom, & all the sides.

$$SA = 2(11 \times 10) + 2(4 \times 10) + 2(4 \times 11)$$

$$= 2(110) + 2(40) + 2(44)$$

$$= 220 + 80 + 88$$

$$SA = 388 \text{ in}^2$$

Surface Area Formulas

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You will be expected to find the surface areas of rectangular prisms, pyramids, cylinders, cones, and spheres. To make your work easier, the GED math formula sheet includes formulas for finding the surface areas of these figures.

Surface Area (SA) Formulas

(p = perimeter of base, B = area of the base,
 h = height, r = radius, s = slant height)

Rectangular/Right Prism	$SA = ph + 2B$
Cylinder	$SA = 2\pi rh + 2\pi r^2$
Square Pyramid	$SA = \frac{1}{2}ps + B$
Cone	$SA = \pi rs + \pi r^2$
Sphere	$SA = 4\pi r^2$

Remember, a formula uses algebra to write a procedure for doing something. These formulas add the areas of the faces of a figure. You don't have to use formulas, but they may save you time.

USING LOGIC

USING A FORMULA TO FIND SURFACE AREA

Example Find the surface area of the pyramid.

Use the formula: $SA = \frac{1}{2}ps + B$

Step 1

Find the facts you need to use the formula.

The slant height s is 4.3 feet.

Find p , the perimeter of the base:

$$3 + 3 + 3 + 3 = 12$$

Find B , the area of the base: $3^2 = 9$

Step 2

Evaluate the formula.

$$\begin{aligned} SA &= \frac{1}{2}ps + B \\ &= \frac{1}{2} \cdot 12 \cdot 4.3 + 9 \\ &= 34.8 \text{ sq ft} \end{aligned}$$

The surface area is 34.8 sq ft.

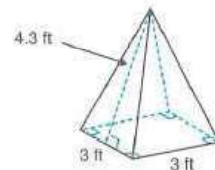
① square base

$$\text{Area} = 3^2 = 9$$

② 4 TRIANGULAR SIDES

$$\text{area} = 4 \left[\frac{1}{2} (3) (4.3) \right]$$

$$= 25.8$$



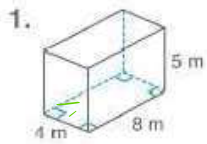
$$SA = 34.8 \text{ Ft}^2$$

∴ LOGIC VS. Memorizing another formula.

Examples

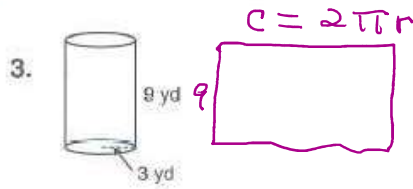
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Find the surface area of each figure. If necessary, round your answer to the nearest tenth square unit.



$$\begin{aligned} SA &= 2(4 \times 5) \\ &+ 2(4 \times 8) \\ &+ 2(8 \times 5) \\ &= 2(20 + 32 + 40) \end{aligned}$$

$$SA = 184 \text{ m}^2$$

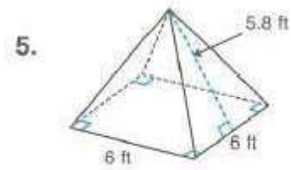


$$\begin{aligned} \text{Area of the 2 Bases} \\ A &= 2(\pi r^2) \\ &= 2(3.14)(3^2) \\ &= 56.52 \end{aligned}$$

$$\begin{aligned} \text{Area of the side} \\ &= h \cdot w \\ &= 9[2(3.14)(3)] \\ &= 169.56 \end{aligned}$$

$$\begin{aligned} SA &= 56.52 + 169.56 \\ &= 226.08 \end{aligned}$$

$$SA = 226.1 \text{ yd}^2$$



$$\begin{aligned} \text{Area of 4 } \Delta \\ &= 4\left(\frac{1}{2}bh\right) \\ &= 4\left(\frac{1}{2}\right)(6)(5.8) \\ &= 69.6 \end{aligned}$$

$$\begin{aligned} \text{Area of Base} \\ A &= s^2 \\ &= 6^2 \\ &= 36 \end{aligned}$$

Area of square
Pyramid

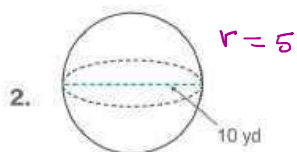
$$A = 69.6 + 36$$

$$A = 105.6 \text{ ft}^2$$

Examples, cont'd...

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Sphere $SA = 4\pi r^2$

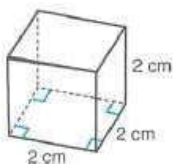


$$SA = 4(3.14)(5^2)$$

$$SA = 314 \text{ yds}^2$$

Derivation of the formula requires calculus.

4.



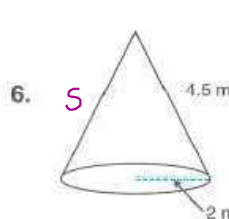
Cube: 6 Faces

$$SA = 6(2^2)$$

$$SA = 24 \text{ cm}^2$$

area of side + Area of Base

Cone $SA = \pi r s + \pi r^2$



$$r = 2$$

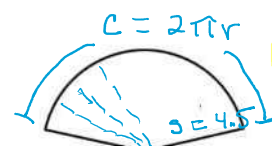
$$s = 4.5$$

$$SA = (3.14)(2)(4.5) + (3.14)(2^2)$$

$$= 3.14(9 + 4)$$

$$= 3.14(13)$$

$$SA = 40.82$$



How does this formula work?

Imagine slicing this sheet into infinitely narrow triangles.

The area of each Δ is $\frac{1}{2}bh$.

The total area is the sum of all the triangles:

$$\frac{1}{2}b_1h + \frac{1}{2}b_2h + \frac{1}{2}b_3h + \dots$$

$$= (\frac{1}{2}h)(\text{sum of lengths of all the bases})$$

$$= \frac{1}{2}h(c)$$

$$= \frac{1}{2}h(2\pi r)$$

$$A = \pi r h$$

And more Examples

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Solve as directed.

7. The illustration shows a truck bed with no top. What is the surface area of the inside of the truck bed?



$$\begin{aligned}
 SA &= 2(8 \times 3) \\
 &+ 2(11 \times 3) \\
 &+ 1(8 \times 11) \\
 &= 2(24) \\
 &+ 2(33) \\
 &+ 88 \\
 &= 48 + 66 + 88
 \end{aligned}$$

$$SA = 202 \text{ sq ft}$$

8. Crunchy Oats cereal comes in a box that measures 8 by 11 by 2 inches. Bran Crisps cereal comes in a box that measures 7 by 10 by 3 inches. (Hint: Make a drawing with labels.)

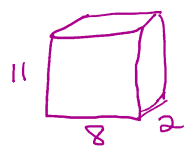
- a. Which box has a larger surface area?

CRUNCHY OATS

- b. Which box has a greater volume?

BRAN CRISPS

Oats



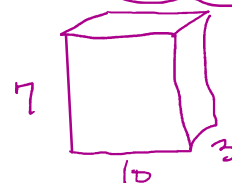
$$\begin{aligned}
 SA &= 2(11 \times 8) \\
 &+ 2(11 \times 2) \\
 &+ 2(8 \times 2) \\
 &= 2[88 + 22 + 16] \\
 &= 2[126]
 \end{aligned}$$

$$SA = 252 \text{ in}^2$$

$$\begin{aligned}
 V &= Lhw \\
 &= 11(8)(2)
 \end{aligned}$$

$$V = 176 \text{ in}^3$$

Crisps



$$\begin{aligned}
 SA &= 2(7 \times 10) \\
 &+ 2(7 \times 3) \\
 &+ 2(10 \times 3) \\
 &= 2[70 + 21 + 30] \\
 &= 2[121]
 \end{aligned}$$

$$SA = 242 \text{ in}^2$$

$$\begin{aligned}
 V &= Lhw \\
 &= 7(10)(3)
 \end{aligned}$$

$$V = 210 \text{ in}^3$$

Problem Solving with Surface Area & Choosing Area, Perimeter, Volume, or Surface Area

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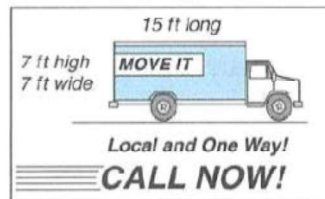
CHOOSING AREA, PERIMETER, VOLUME, OR SURFACE AREA

You know how to find the perimeter, area, volume, and surface area of various figures. But knowing which to solve for takes practice.

Remember: **Perimeter** is the distance around a flat figure. **Area** is the space inside a flat figure. **Volume** is the space inside a three-dimensional figure. **Surface area** is the sum of the areas of the faces of a three-dimensional figure.

For each problem, select whether to solve for perimeter, area, volume, or surface area. Then solve the problem. Round to the nearest whole unit if necessary.

1. How many cubic feet does the moving truck shown in the ad hold?



$$V = LWH \\ = 7(7)(15)$$

Solve for:

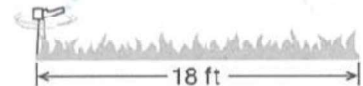
perimeter
area
<u>volume</u>
surface area

Solution: $V = 735 \text{ ft}^3$

2. A revolving sprinkler sprays a lawn for a distance of 18 feet. How much ground is covered by the sprinkler as it makes 1 complete revolution?

$$A = \pi r^2 \\ = (3.14)(18^2)$$

CIRCLE



Solve for:

perimeter
<u>area</u>
volume
surface area

Solution: $A = 1017 \text{ ft}^2$

More Examples

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3. How many feet of chrome edging are needed to finish a rectangular kitchen table?



edging has thickness

Solve for:

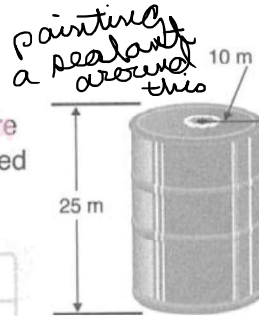
perimeter
area
volume
surface area

$$P = 2L + 2W \\ = 2(2.5) + 2(4) \\ = 5 + 8$$

Solution: $P = 13\text{ ft}$

width = perimeter on side
length = same thing
 $\therefore$ width

4. To prevent leakage, an oil storage tank is coated with a sealant. How many square meters will need to be coated to cover the container?



Solve for:

perimeter
area
volume
surface area

Solution: 2198 m^2

Surface Area of the Can

Top Area = πr^2

Bottom Area = πr^2

Side Area = $\pi d h$

$C = \pi d$
 h

$r = 10$

$d = 20$

$C = \pi d = 2\pi r = 20\pi$

$h = 25$

6. A public school wants to install rubber padding over a rectangular playground space that measures 22 feet by 15 feet. The padding comes in square tiles that measure 1 foot by 1 foot. How many tiles are needed?

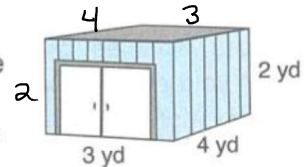
Solve for:

perimeter
area
volume
surface area

$$A = 22 \times 15 \\ = (20 \times 15) + (2 \times 15) \\ = 300 + 30 \\ A = 330\text{ ft}^2$$

Solution: 330 Tiles

7. A storage unit has the dimensions shown in the drawing. If the top and four sides of the unit are repainted, how many square yards will be painted?



Solve for:

perimeter
area
volume
surface area

Solution: 40 yds^2

Top 4×3
2 Short sides 3×2
2 Long sides 4×2
 $SA = (4)(3) + 2(3)(2) + 2(4)(2)$
 $= 12 + 12 + 16$
 $SA = 40\text{ sq yds}$

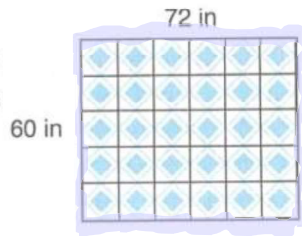
$$SA = 2\pi r^2 + 2\pi r h \\ = \pi (2r^2 + 2rh) \\ = \pi [2(10^2) + 2(10)(25)] \\ = \pi (200 + 500) \\ = 700\pi$$

$SA = 2198\text{ m}^2$

More Examples, cont'd...

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5. Marisol wants to put a **satin border** on a quilt she made. The drawing shows the dimensions of the quilt. **How many inches long is the border?**



Solve for:

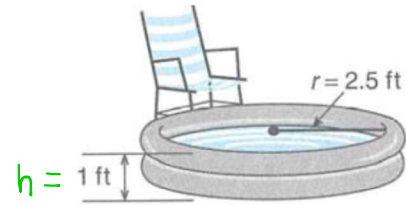
perimeter
area
volume
surface area

$$\begin{aligned}
 P &= 2L + 2W \\
 &= 2(4 + 6) \\
 &= 2(10) \\
 &= 20
 \end{aligned}$$

Solution:

244 in

8. How many **cubic feet** of water will the circular wading pool hold?



Solve for:

perimeter
area
volume
surface area

$$\begin{aligned}
 V &= \pi r^2 h \\
 &= \pi (2.5)^2 (1) \\
 &= 6.25\pi \\
 V &= 19.625
 \end{aligned}$$

Solution:

20 ft³